Topic	**JAVASCRIPT STATEMENTS**	
Class Description	**Data Structures are one of the most applied concepts of software engineering. Kids understand the concept of one of the most common data structures - Array and apply associated methods.**	
Class	**ADV-C78**	
Class time	**55 mins**	
Goal	● Implement Arrays to methods.	
Resources Required	● Teacher Resources ○ Use gmail login credentials ○ Earphone with mic ○ Notepad and Pen ● Student Resources ○ Use gmail login credentials ○ Earphone with mic (optional) ○ Notepad and Pen	
Class structure	**Warm-Up** **Teacher-Led Activity** **Student-Led Activity** **Wrap Up** **Project Pointers and Cues**	**5 Mins** **10 Mins** **30 Mins** **5 Mins** **5 Mins**
Start the video call from H2H		
WARM UP SESSION - 5 mins		
Teacher starts slideshow from slides 1 to 23. Refer to speaker notes and follow the instructions on each slide.		
TEACHER ACTIVITY - 10 mins		
Teacher Initiates Screen Share		

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Say	Do
Let's understand now what an array is.	Run Teacher-Activity 1 and show the output on the browser as well as on the console.

Explanation of arrays

● An array is a variable, which can hold more than one value at a time. ● If you have a list of items (for example: A list of friend names), storing the names in a single variable could look like this:

```
var friend1 = "Ram";
var friend2 = "Rohit";
var friend3 = "Rocky";
```

● What if you have not just 3 friends, but 200 friends? Then solution is an array! ● An array can hold many values under a single name, and you can access the values by referring to an index number.

For example: var friends = ["Ram", "Rohit", "Rocky"];

Now I will explain to you the code and output of Teacher-Activity 1.

We already know HTML basic code so we do not want to waste time writing the same. So HTML code is already pre-written. As a brief - we will have **h4 tags** and we are updating these tags from javascript using the id of the element.
For example: Refer to Teacher-Activity-1 Image 1

We will start adding Javascript code to the pre-written HTML code.

Let's begin.

1. How to declare the array?

- First, we define a variable using **var** keyword, followed by the name that we want.
- Then we put all the values we want inside the square brackets **[ ].** ○ For strings the values should be in double quotes and every value should be separated using commas.

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- So we are defining the array as fruits and putting all the fruits we want inside this fruits array.

- Now console the array.
  - We will write the variable name inside the console.log, because we want to print the whole array on the console.

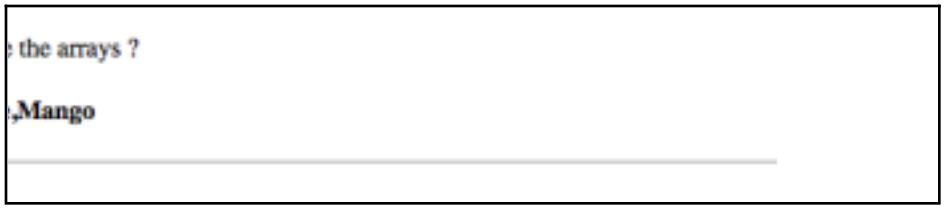
```
var fruits = ["Apple", "Orange", "Mango"];
console.log(fruits);
```

- Output on console screen:

- Now that we are sure of the fruits array we declared, so we will print this array on webpage using id of the HTML element.

```
document.getElementById("declaration_of_arrays").innerHTML = fruits;
```

- Output on the web screen:



2. How to fetch the first element from the array?

- You can access an array element by referring to the index number.
  - This statement accesses the value of the first element in fruits: **fruits[0]**
  - Array indexes start with 0. **[0]** is the first element. **[1]** is the second element, and so on..

- First we will declare a variable first_fruit and store fruits[0] inside it.

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  - **fruits[0]** will get the first fruit from the array which is Apple and store it inside the variable first_fruit.

- Console this variable

- Output on console.log:

- Now we will print this first element from the array on the HTML element using its id.

- Output on the web screen:

3. How to fetch the second element of the array
   - We will declare another variable second_fruit and store fruits[1] inside it. ○ **fruits[1]** will get the second fruit from the array which is Orange and store inside the variable second_fruit.

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- Console this variable.

- Output on console.log:

- Now we will print this first element from the array on the HTML element using its id.

- Output on the web screen:

4. Same way, fetch the third element of the array.

   - Code:

5. How to fetch the length of the array?

   - Length of an array is the number of elements in the array.
     - For example: In the following code, the length of the fruits array is 3, because it has 3 elements.

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- We declare a variable **length_of_the_array** and store **fruits.length** ○ This will give 3, because the fruits array has three elements.

- Then we will console the variable to see if we are getting the right output.

- Output on console screen:

- Now we will print the output on the HTML element.

- Output of web screen:

6. How to loop through the array?

- We will loop through the array using **for loop.**
  - ○ Looping through arrays is done because if we have an array that has 200 elements, it is impossible to display each element using the method we discussed in points 2, 3 and 4.
  - ○ So for making the code easy and short we loop through the array. ● First we will declare an empty variable **array_value.**

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- We will use for loop:
    - Set **i=0;** Set it to 0 because the array's first index is 0.
        - Set the termination condition to lesser than **fruits.length,** because we want the loop to stop when it reaches the last element of the array.
    - Increase the value of **variable i** by 1, so one by one we get the elements from the array.
    - Console the elements of the array one by one.
    - Update the variable **array_value** by the element of the array plus a **br tag.**

- **First i = 0**
  console.log(fruits[i])
  means console.log(fruits[0]) // because i =0
  fruits[0] , means Apple // As we discussed in point 2 that fruits[0] = Apple

- **array_value =** array_value + fruits[i] + “<br>”;
  means **array_value =** “ ” + Apple + “<br>”;

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● Now i will be incremented by 1 because we have used **i++** in for loop **So now i = 1** console.log(fruits[i]) means console.log(fruits[1]) // because i =1 fruits[1] , means Orange // As we discussed in point 3 that fruits[1] = Orange

● **array_value =** array_value + fruits[i] + “<br>”;
means **array_value =** Apple
 + Orange + “

- Again i will be incremented by 1 because we have use **i++** in for loop
  So now i = 2
  console.log(fruits[i])

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means console.log(fruits[2]) // because i =2
fruits[2] , means Mango // As we discussed in point 4 that fruits[2] = Mango

- **array_value =** array_value + fruits[i] + "<br>";
  means **array_value =** Apple
Orange
 + Mango + "

- Again i will be incremented by 1 because we have use i++ in for loop.
- **So now because i = 3:**
    - The loop will stop. This is because the termination condition was

when it reaches the last element of the array loop should stop.

- So the output on console will be:

- After the loop we print the output on the HTML element.

- So the HTML element with id=**loop_the_array** will be updated with the value of the variable **array_value**.

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- And the output on the web screen will be:

7. Now we will add an element to the array.

- **push()** function will add elements at the end of the array.
    - Before using push function the array - Apple, Orange, Mango
    - After using push function the array - Apple, Orange, Mango, Lemon

- Now console the array.

- Output on console screen:

- Printing the output on the HTML element.

- Output on web:

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8. Add one more element in the same way as shown above.

- Code:

- Output on console:

- Output on web:

9. Declare one more array and display it the same way we did for fruits.

- Code:

- Output on console:

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- Output on web:

10.Sort the array

Now we will learn how to sort the array, and this sorting of array is alphabetical.

- **sort()** function is used to sort the array.
  - The following will sort the alphabets array.

- Then we will console the alphabets array.

Output on console:

Then we will display on the HTML element.

- Output on web:

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11. Reverse the array
- Now we will learn how to reverse the array.
- **reverse()** function is used to reverse the array.
  - The following will reverse the alphabets array.

- Then we will console the alphabets array.

- Output on console:

- Then we will display on the HTML element.

- Output on web:

12. Declare one more array and display it the same way as we did for fruits.

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- Code:

- Output on console:

- Output on web:

13. Now we will find the maximum number from the numbers array.
    - **Math.max.apply(Math, array_name)** function is used to find the maximum number from the array.
    - We will declare a variable **maximum_number** and store the function of finding the **max** value from the array.
    - We will use **Math.max.apply(Math, array_name)** pass the array name which is numbers.

- Then we will console the variable **maximum_number.**

- Output on console:

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- Then we will display on HTML element.

- Output on web:

14. Now we will find the minimum number from the numbers array.
    - **Math.min.apply(Math, array_name)** function is used to find the minimum number from the array.
    - We will declare a variable **minimum_number** and store the function of finding the **min** value from the array.
    - We will use **Math.min.apply(Math, array_name)** pass the array name which is numbers.

- Then we will console the variable **minimum_number.**

- Output on console:

- Then we will display on HTML element:

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- Output on web:

Teacher Stops Screen Share
STUDENT ACTIVITY - 30 mins
Now it is your turn.
● **Ask Student to press ESC key to come back to panel** ● **Guide Student to start Screen Share** ● **Teacher gets into Fullscreen**
Student Initiates Screen Share

Say	**Do**

Now let the student do the javascript code as explained above. The student should download the HTML file present in Student-Activity-1 and put it in his local system and start JS coding. The HTML part is given, the student needs to do only the javascript part. ***NOTE - the HTML element id to be used in**	**Student Activity 1** **SOURCE CODE** **Student Activity 2** **CODE DIAGRAM** **Student Activity 3** **JS ARRAY REFERENCE** **Student Activity 4** **JS LOOP ARRAY REFERENCE**

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document.getElementById().innerHTM L, is mentioned in the comments For example:	**Student Activity 5** **JS PUSH REFERENCE** **Student Activity 6** **REFERENCE FOR HOSTING A WEBSITE ON GITHUB**

Now let's host your website using GitHub. **The steps to host the website is mentioned in Student-Activity-6, copy the github and paste it in the Submit Link Field on the student panel and click Submit Button.** **Test your webapp using this link on your Phone.**	Encourage and help the student if the student gets stuck. **NOTE TEACHERS -** **First ask the student to host the website, the steps to host the website is mentioned in Student-Activity-6, and copy the github link and paste it in the Submit Link Field on the student panel and click Submit Button.** **Test the webapp using this link on your laptop.**
Teacher Guides Student to Stop Screen Share	
Teacher Initiates Screen Share	
WRAP UP SESSION - 5 mins	

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Teacher starts slideshow from slides 24 to 27. Refer to speaker notes and follow the instructions on each slide.
PROJECT POINTERS AND CUES - 5 mins
Teacher starts slideshow at slide 28. Refer to speaker notes and follow the instructions on each slide.

Say	Do

Project Name: FAMILY BOOK **Goal of the Project:** In Class 78, we learned about arrays. We also learned to write JavaScript code inside the Html document itself. In this project, you have to build a family book web app. **Story:** Let's make a **family book** web app, where you can store all the names and photos of your family members. In this app, if you click on the Next button, the image and name of the next family member are displayed. I am very excited to see your Family book and I know you will do really well. Good Luck!	

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Teacher Stops Screen Share
Teacher Clicks
ADDITIONAL ACTIVITY
Teacher Initiates Screen Share

Teacher starts slideshow from slides 29 to 30.	
STUDENT ADDITIONAL ACTIVITY	
Student Initiates Screen Share	
Say	**Do**
As you saw there was additional activity in teacher-activity-1 so now I will explain the code here. 1. Now we will add an element at the beginning of the array. ● **unshift()** function will add element in the beginning of the array ○ Before using unshift function the array - Apple, Orange, Mango, Lemon, Cheery ○ After using unshift function the array - watermelon, Apple, Orange, Mango, Lemon	**Guide the student in solving the additional activity.**

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- Now console the array.

- Output on console screen:

- Print on HTML element.

- Output on web:

2. Add one more element at the beginning of the array the same way we added watermelon.

- Code:

- Console output:

- Output on web:

3. How to remove an element from the beginning of the array?

- **shift()** function will remove an element from the

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beginning of the array.

- Before using shift function the array - strawberry, watermelon, Apple, Orange, Mango, Lemon, Cheery
- After using shift function the array - watermelon, Apple, Orange, Mango, Lemon, Cheery

- Now console the array.

- Output on console screen:

- Print on HTML element.

- Output on web:

4. How to remove an element from the end of the array?
   - **pop()** function will remove element from the end of the array
     - Before using pop

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function the array - watermelon, Apple, Orange, Mango, Lemon, Cheery
    - After using pop function the array - watermelon, Apple, Orange, Mango, Lemon

- Now console the array.

- Output on console screen:

- Print on HTML element.

- Output on web:

5. How to merge two arrays?
    - Define two arrays.
        - First world country array:

        - Second world

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country array:

- Now define a variable **all_countries** and assign it the function **concat()**.
  - **concat function** is used to merge two arrays.
  - First the array name followed by the **concat function** and inside that the other array which needs to be joined.

- Now console the array.

- Output on console screen:

- Print on HTML element.

- Output on web:

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Activity	Activity Name	Links
Teacher Activity 1	ARRAY WEBSITE	https://mahdihat791.github.io/v2/ADV-C78/
Teacher Activity 2	CODE DIAGRAM	https://docs.google.com/document/d/e/2PACX-1vStvXT9U0KUyeu1ViqVen6J2orL0WOBdu7fUHVRWbF3Gvs8kMQOxLtCHGaLi c7z9giSNMw07EiGDm8y/pub
Teacher Activity 3	SOURCE CODE	https://drive.google.com/open?id=1ZmdbM 4LiXQpAD3LcEKThNhacqybO0TsU
Student Activity 1	SOURCE CODE	https://drive.google.com/drive/folders/1Ur2RXbPVU5SwP1NiTmRYzVQHHMnEGCU _?usp=sharing
Student Activity 2	CODE DIAGRAM	https://docs.google.com/document/d/e/2PACX-1vStvXT9U0KUyeu1ViqVen6J2orL0WOBdu7fUHVRWbF3Gvs8kMQOxLtCHGaLi c7z9giSNMw07EiGDm8y/pub
Student Activity 3	JS ARRAY REFERENCE	https://www.w3schools.com/js/js_arrays.as p
Student Activity 4	JS LOOP ARRAY REFERENCE	https://www.w3schools.com/js/js_loop_for. asp
Student Activity 5	JS PUSH REFERENCE	https://www.w3schools.com/jsref/jsref_pus h.asp

Student Activity 6	REFERENCE FOR HOSTING A WEBSITE	https://docs.google.com/document/d/e/2PACX-1vQ-vDFJOR8yO_oQYzNCfE_QbXm56bvvowINBmXQIvuhZHhnC4rMSXMhUOb Mk9H1IYtQLfjl-3pjplHU/pub
Project Solution	FAMILY BOOK	https://snehalbsawant.github.io/ADVC78/

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Teacher Reference Visual aid link	Visual aid link	https://s3-whjr-curriculum-uploads.whjr.online/762e 8a57-cf55-4eca-9081-c94f5b3eb8be.html
Teacher Reference In-class quiz	In-class quiz	https://s3-whjr-curriculum-uploads.whjr.online/fafee5f0-60f4-4fd5-8776-cf3af5aa1943. pdf

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